

ABSTRACT

Heart disease is one of the leading causes of death worldwide. Early prediction of heart disease helps doctors provide timely treatment and improve patient outcomes.

This project presents a Heart Attack Prediction System using Machine Learning techniques. The system analyses patient medical parameters such as age, blood pressure, cholesterol level, chest pain type, fasting blood sugar, electrocardiographic results, maximum heart rate, exercise-induced angina, oldpeak, ST slope, number of major vessels, and thalassemia.

The project includes data preprocessing, exploratory data analysis (EDA), visualization, feature correlation analysis, and machine learning model development. Multiple classification algorithms such as Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbours (KNN), Support Vector Machine (SVM), and Gaussian Naive Bayes were trained and evaluated.

Among all the models, Decision Tree and Random Forest achieved the highest accuracy on the test dataset. The developed system demonstrates that machine learning can effectively assist in predicting heart disease using patient clinical data.

Keywords: Machine Learning, Heart Disease Prediction, Data Analysis, Classification, Random Forest, Decision Tree, Healthcare Analytics.

CHAPTER 1 INTRODUCTION

Heart disease is one of the most serious health problems affecting millions of people around the world. Detecting heart disease at an early stage is extremely important because timely diagnosis can significantly reduce mortality and improve treatment outcomes.

With the rapid growth of Artificial Intelligence and Machine Learning, healthcare organizations are increasingly using predictive models to assist doctors in making faster and more accurate decisions. Machine learning algorithms analyse historical medical data and identify hidden patterns that may not be easily recognized by humans.

This project develops a Heart Attack Prediction System using machine learning algorithms. The model predicts whether a patient is likely to have heart disease based on various medical attributes including age, blood pressure, cholesterol level, chest pain type, electrocardiographic results, fasting blood sugar, exercise-induced angina, maximum heart rate, oldpeak, ST slope, number of major vessels, and thalassemia.

The project also performs extensive Exploratory Data Analysis (EDA) to understand the dataset through statistical analysis and visualizations. Several machine learning algorithms are trained and compared to identify the most accurate prediction model.

CHAPTER 2 LITERATURE SURVEY

2.1 Introduction

Heart disease prediction has become an important area of research in the field of healthcare and machine learning. Researchers have developed various predictive models to assist medical professionals in diagnosing heart disease at an early stage. These models analyse patient health records and identify patterns that indicate the risk of heart disease.

Machine learning techniques have gained popularity because they provide fast, accurate, and automated predictions compared to traditional diagnosis methods.

2.2 Existing System

Traditional heart disease diagnosis depends mainly on clinical examinations, laboratory tests, ECG reports, and the experience of medical professionals.

Although these methods are effective, they have several limitations:

- Time-consuming diagnosis process.
- Human errors due to manual interpretation.
- Difficulty in analysing large amounts of patient data.
- Higher healthcare costs.

2.3 Proposed System

The proposed system uses Machine Learning algorithms to predict the likelihood of heart disease based on patient clinical data.

The system performs:

- Data preprocessing
- Exploratory Data Analysis (EDA)
- Feature selection

- Machine learning model training
- Model comparison
- Performance evaluation

The prediction generated by the system can support healthcare professionals in making informed clinical decisions.

2.4 Advantages of Machine Learning in Healthcare

- Early disease detection
- Improved diagnostic accuracy
- Faster decision making
- Reduced human errors
- Cost-effective healthcare
- Ability to analyse large medical datasets

2.5 Literature Summary

Previous studies demonstrate that machine learning algorithms such as Decision Tree, Random Forest, Logistic Regression, Support Vector Machine, and K-Nearest Neighbours achieve promising accuracy in predicting heart disease.

Among these algorithms, ensemble learning methods like Random Forest often provide higher prediction accuracy due to their ability to reduce overfitting and improve generalization.

CHAPTER 3 PROBLEM STATEMENT

Heart disease remains one of the leading causes of mortality worldwide. Early identification of patients at risk is essential for timely treatment and prevention of serious complications. However, traditional diagnosis methods require extensive medical examinations and expert interpretation.

The objective of this project is to develop a machine learning-based prediction system that analyses patient clinical data and predicts whether a patient is likely to suffer from heart disease. The proposed system aims to improve prediction accuracy while reducing diagnosis time and assisting healthcare professionals in decision-making.

CHAPTER 4 OBJECTIVES

The main objectives of the project are:

- To study heart disease and its major risk factors.
- To collect and analyse a heart disease dataset.
- To perform data preprocessing and cleaning.
- To perform Exploratory Data Analysis (EDA).
- To visualize important medical features using various graphs.
- To train multiple machine learning algorithms.
- To compare model performance using evaluation metrics.
- To identify the most accurate prediction model.
- To assist healthcare professionals in early heart disease prediction.

CHAPTER 5 DATASET DESCRIPTION

5.1 Dataset Overview

The project uses a publicly available Heart Disease Dataset containing **1025 patient records**.

Each record represents one patient and contains clinical information that may contribute to heart disease diagnosis.

The dataset contains **14 attributes**, including the target variable.

5.2 Dataset Features

Attribute Description

Age	Age of patient
Sex	Gender (Male/Female)
CP	Chest Pain Type
Trtbps	Resting Blood Pressure
Chol	Cholesterol
FBS	Fasting Blood Sugar
Rest ECG	Resting Electrocardiographic Results
Thalach	Maximum Heart Rate Achieved
Exang	Exercise Induced Angina
Oldpeak	ST Depression
Slope	Slope of Peak Exercise ST Segment

Attribute Description

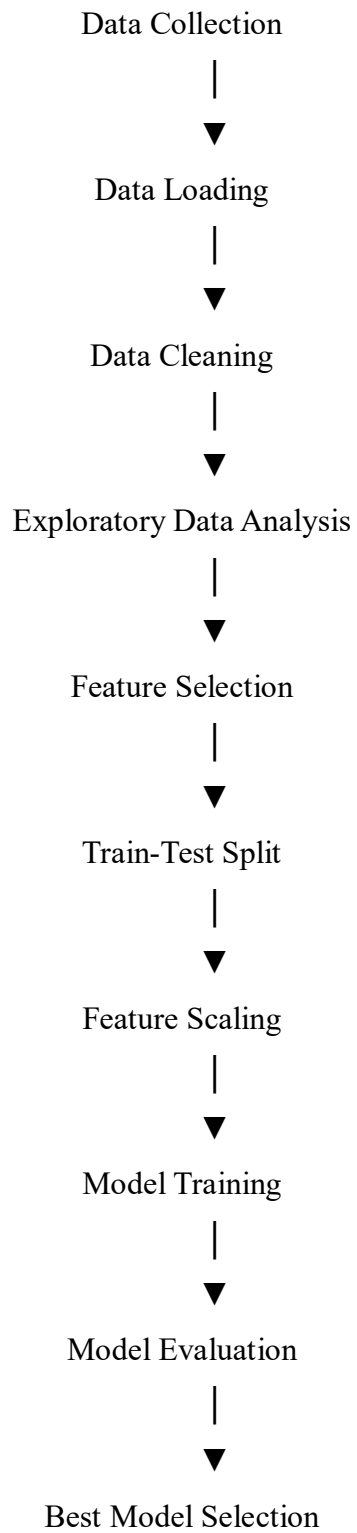
CA	Number of Major Vessels
Thal	Thalassemia
Target	Heart Disease Prediction

5.3 Dataset Statistics

- Total Records: **1025**
- Number of Features: **13**
- Target Variable: **1**
- Missing Values: **0**
- Numerical Variables: **5**
- Categorical Variables: **9**

CHAPTER 6 METHODOLOGY

The project follows the standard Machine Learning pipeline.



6.1 Data Collection

The Heart Disease Dataset was collected from a publicly available repository containing patient clinical records.

6.2 Data Preprocessing

The preprocessing stage included:

- Checking missing values
- Statistical summary
- Renaming column names
- Identifying numerical and categorical variables
- Data visualization
- Feature scaling
- Dataset splitting

6.3 Exploratory Data Analysis

EDA was performed using:

- Histograms
- Box Plots
- Pie Charts
- Count Plots
- Correlation Heatmap

The purpose of EDA was to understand feature distributions, identify outliers, and discover relationships between variables.

6.4 Machine Learning Models

The following algorithms were implemented:

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbours
- Support Vector Machine
- Gaussian Naive Bayes

6.5 Model Evaluation

Performance was evaluated using:

- Accuracy Score
- Confusion Matrix
- Classification Report

The best-performing model was selected based on overall prediction accuracy.

CHAPTER 7 EXPLORATORY DATA ANALYSIS (EDA)

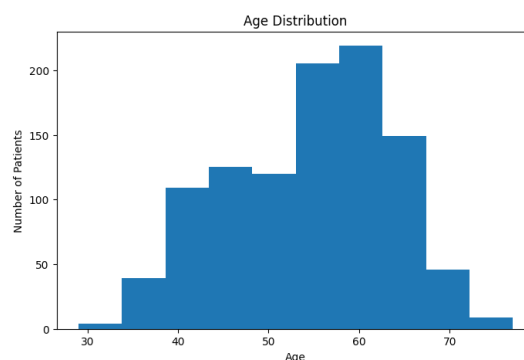
7.1 Introduction

Exploratory Data Analysis (EDA) is the process of analysing and visualizing data to understand its structure, identify patterns, detect outliers, and discover relationships among variables before building machine learning models.

In this project, EDA was performed using histograms, box plots, pie charts, count plots, and a correlation heatmap.

7.2 Histogram Analysis

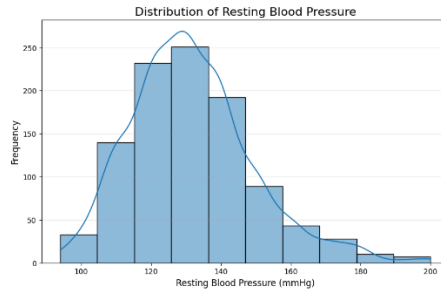
7.2.1 Age Distribution



Interpretation

The histogram shows that most patients are between **50 and 65 years** of age. The highest concentration of patients lies around 58–60 years. Very few patients are below 35 years or above 75 years.

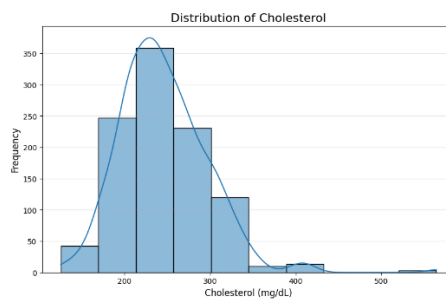
7.2.2 Resting Blood Pressure Distribution



Interpretation

Most patients have resting blood pressure values between **120 and 140 mmHg**. The distribution is slightly right-skewed, indicating a few patients have unusually high blood pressure.

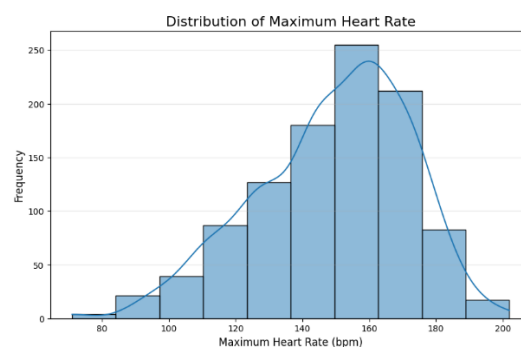
7.2.3 Cholesterol Distribution



Interpretation

The cholesterol distribution is positively skewed. Most patients have cholesterol levels between **200 and 300 mg/dL**, while a few patients have very high cholesterol values exceeding **400 mg/dL**.

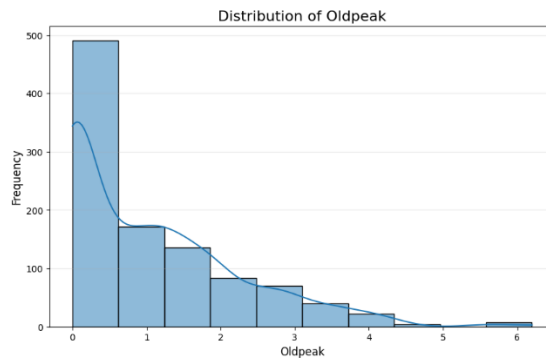
7.2.4 Maximum Heart Rate Distribution



Interpretation

Most patients achieved a maximum heart rate between **140 and 175 bpm**. The distribution is approximately normal with a slight left skew.

7.2.5 Oldpeak Distribution

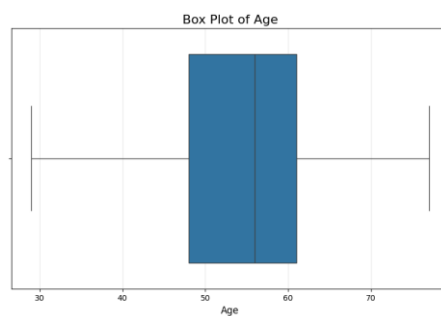


Interpretation

The Oldpeak variable is highly right-skewed. Most patients have values below **2.0**, while only a few patients have higher values indicating severe ST depression.

7.3 Box Plot Analysis

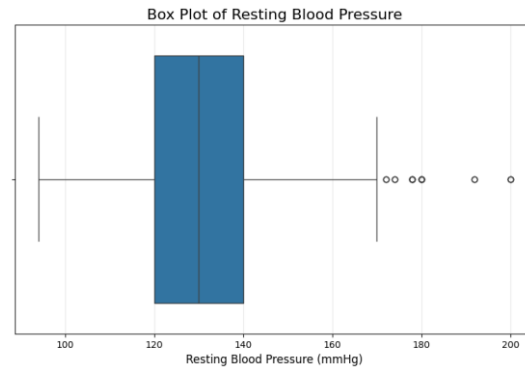
7.3.1 Age Box Plot



Interpretation

No significant outliers are present in the Age variable. The median age is approximately **56** years.

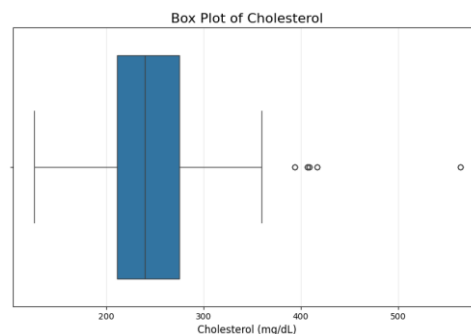
7.3.2 Resting Blood Pressure Box Plot



Interpretation

Several high-value outliers are observed above **170 mmHg**, indicating patients with hypertension.

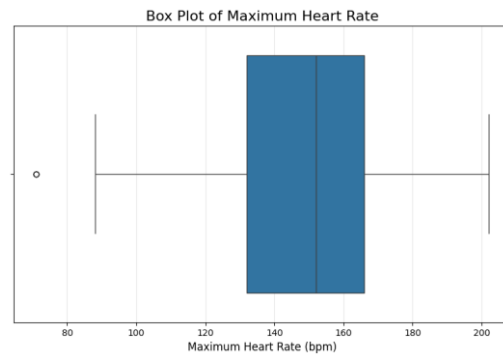
7.3.3 Cholesterol Box Plot



Interpretation

The cholesterol variable contains several extreme outliers above **390 mg/dL**, representing patients with unusually high cholesterol levels.

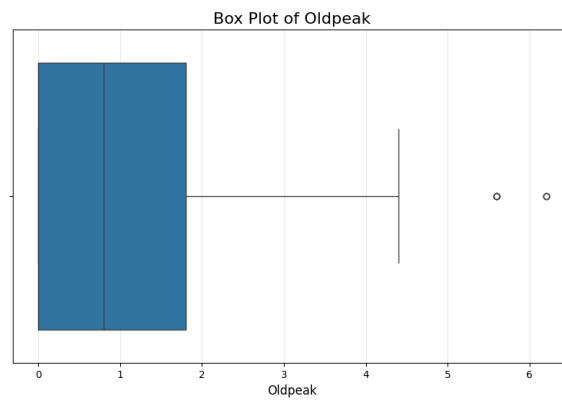
7.3.4 Maximum Heart Rate Box Plot



Interpretation

One low-value outlier is observed around **72 bpm**. The remaining values are evenly distributed.

7.3.5 Oldpeak Box Plot

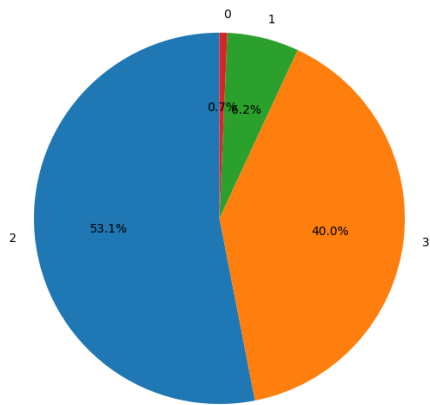


Interpretation

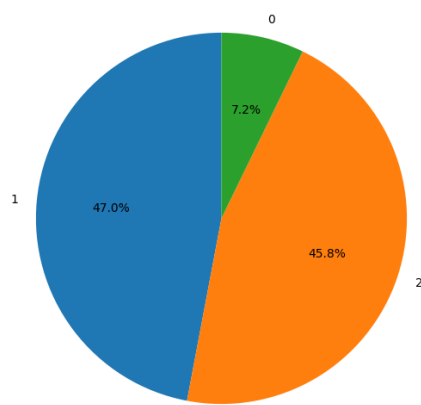
A few high-value outliers are present above **5.5**, indicating patients with severe ST depression.

7.4 Pie Chart Analysis

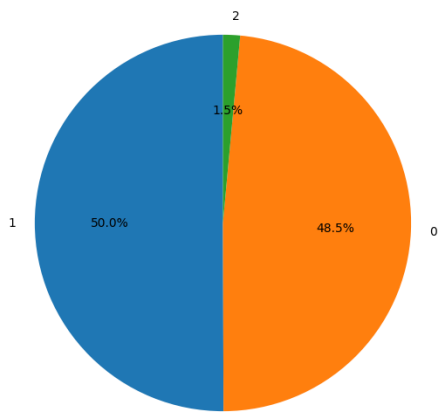
Thalassemia Distribution



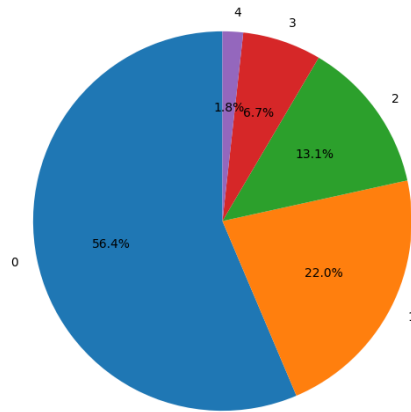
ST Slope Distribution



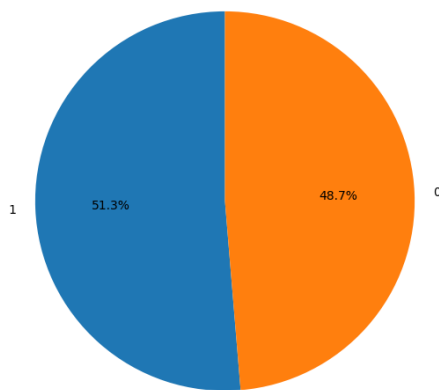
Resting ECG Distribution



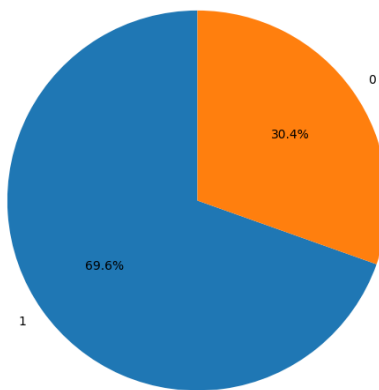
Number of Major Vessels Distribution



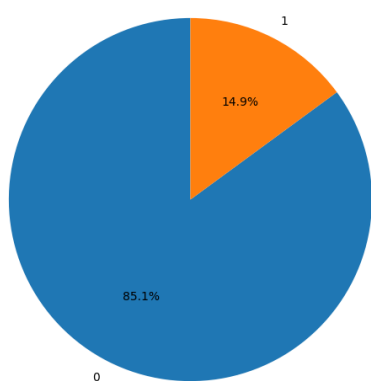
Heart Disease Distribution



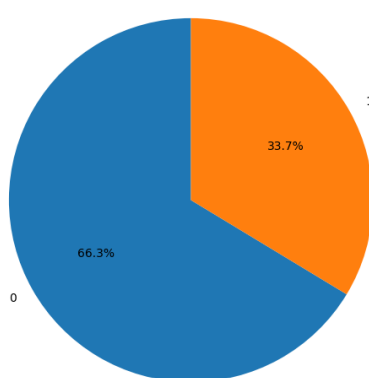
Gender Distribution



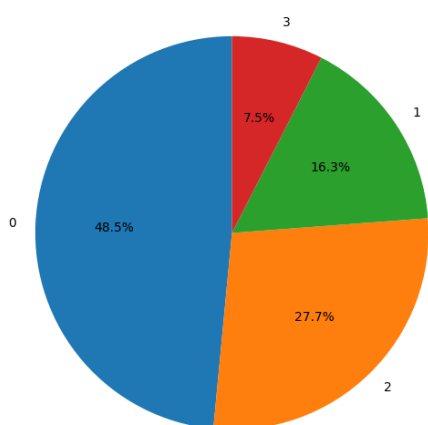
Fasting Blood Sugar Distribution



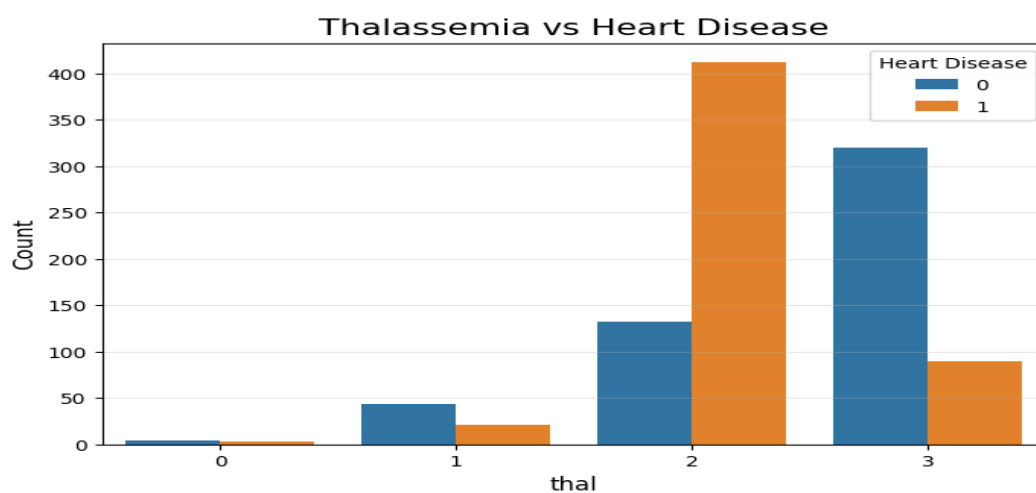
Exercise-Induced Angina Distribution

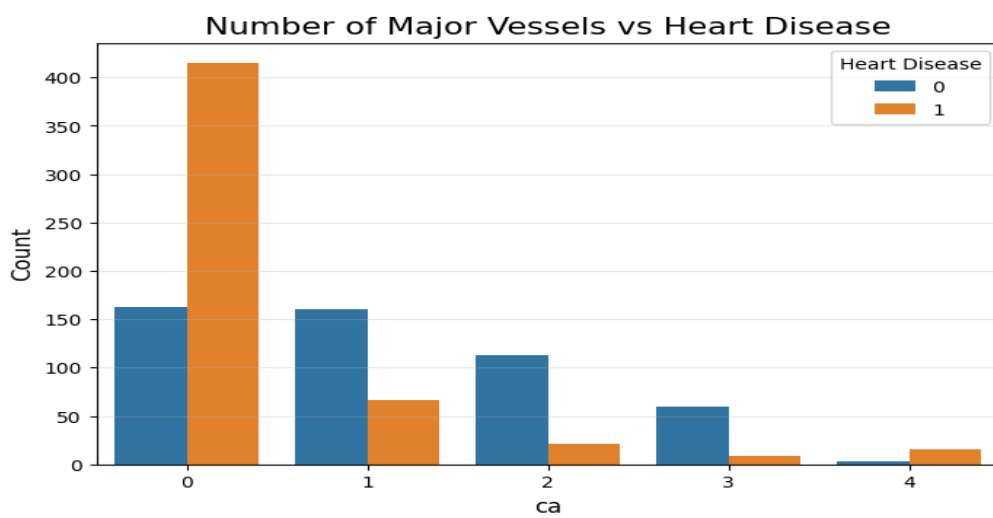
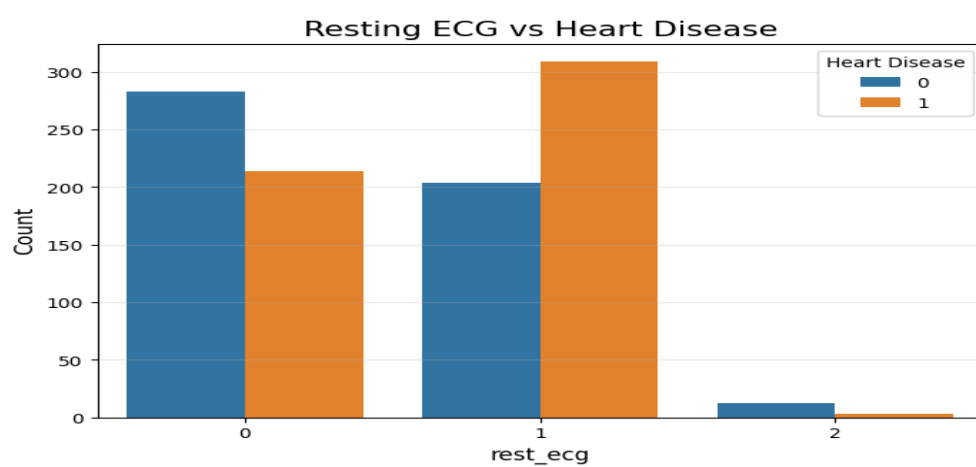
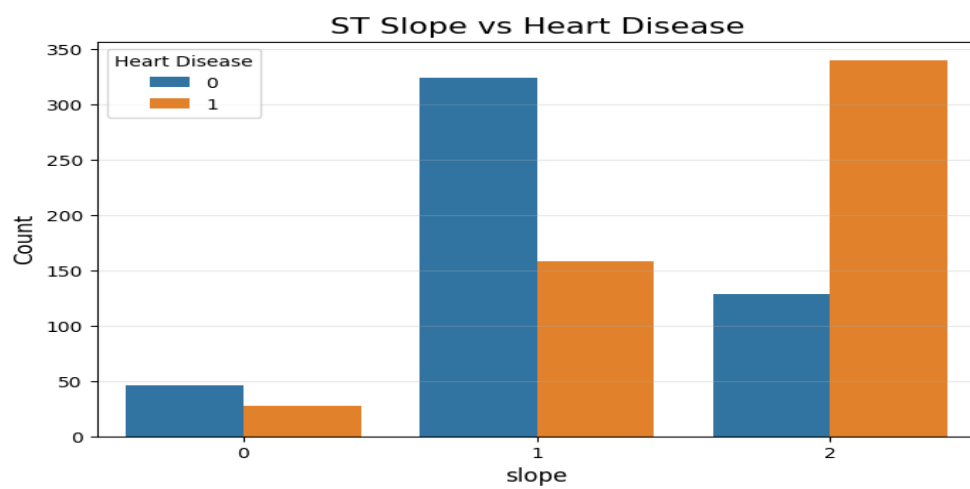


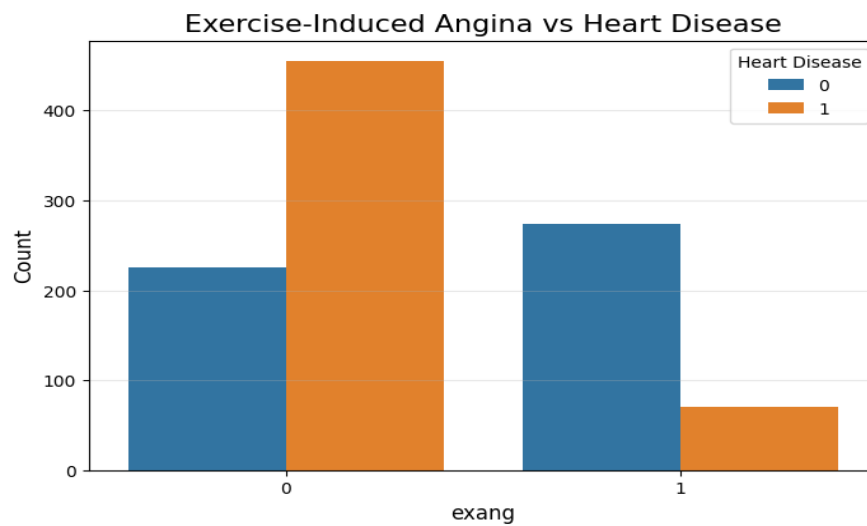
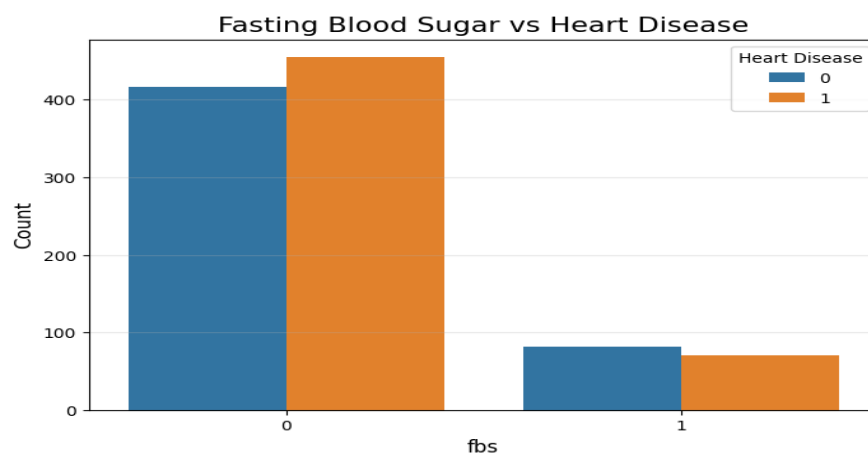
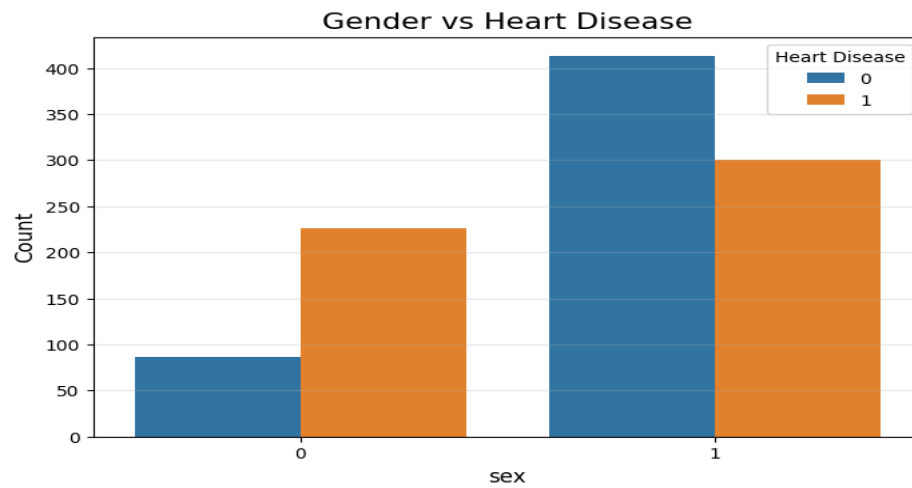
Chest Pain Type Distribution

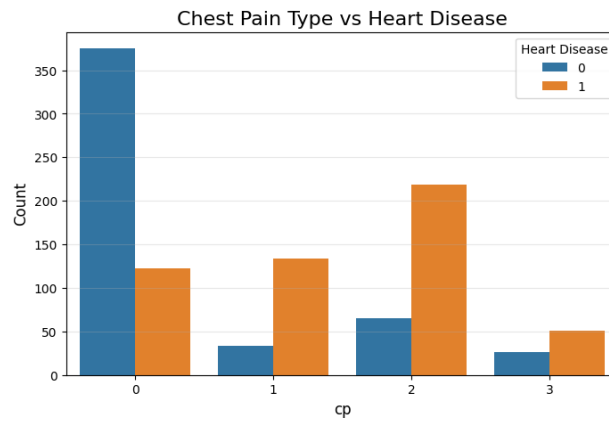


7.5 Count Plot Analysis

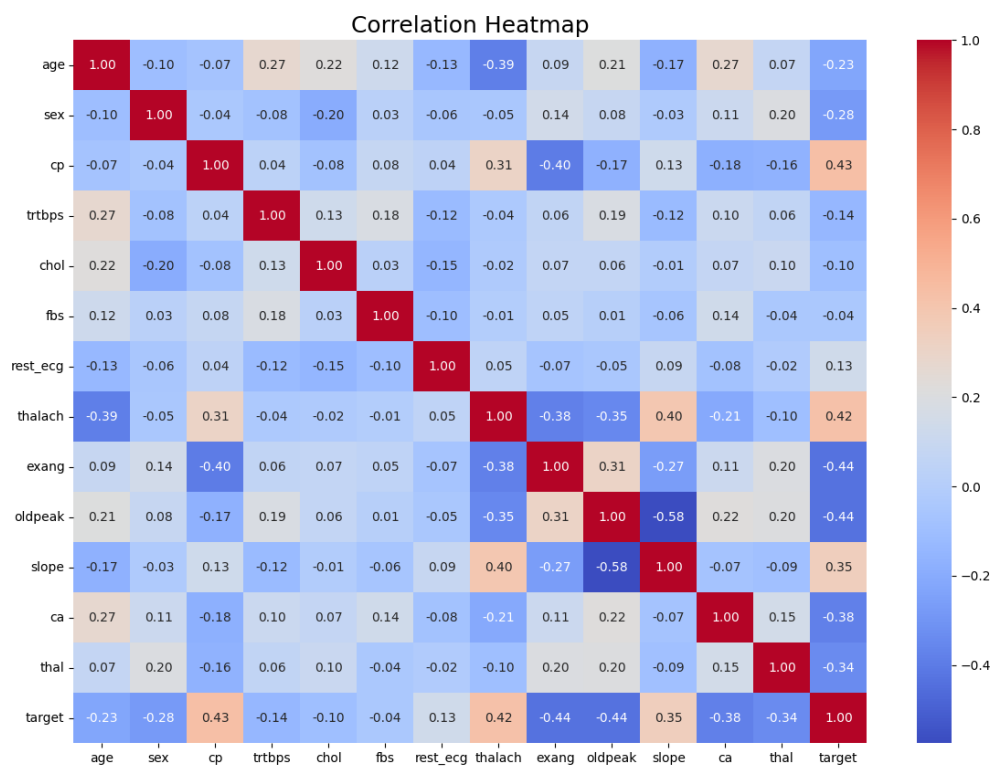








7.6 Correlation Analysis



Interpretation

The correlation heatmap shows the relationship between all features.

- Chest pain (cp) has the strongest positive correlation with heart disease.
- Maximum heart rate (thalach) and ST slope (slope) also show positive relationships.
- Exercise-induced angina (exang), Oldpeak, Number of Major Vessels (ca), and Thalassemia (thal) exhibit negative correlations with the target variable.

The heatmap helps identify important features for model building.

CHAPTER 8 MACHINE LEARNING MODELS

The following supervised classification algorithms were implemented:

8.1 Logistic Regression

Logistic Regression is a linear classification algorithm that estimates the probability of a binary outcome using the logistic function. It serves as a simple and interpretable baseline model.

8.2 Decision Tree

Decision Tree classifies data by recursively splitting it into branches based on feature values. It is easy to interpret and can model complex decision boundaries.

8.3 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

8.4 K-Nearest Neighbours (KNN)

KNN classifies a new data point based on the majority class among its nearest neighbours using distance-based similarity.

8.5 Support Vector Machine (SVM)

SVM identifies the optimal hyperplane that maximizes the separation between classes, making it effective for binary classification.

8.6 Gaussian Naive Bayes

Gaussian Naive Bayes applies Bayes' theorem under the assumption that features follow a Gaussian distribution and are conditionally independent.

CHAPTER 9 RESULTS AND DISCUSSION

Model Performance

Model	Accuracy
Decision Tree	99%
Random Forest	99%
SVM	89%
KNN	84%
Logistic Regression	80%
Gaussian Naive Bayes	80%

(Insert Accuracy Comparison Bar Chart Here)

The Decision Tree and Random Forest models achieved the highest accuracy among all algorithms tested. These models demonstrated excellent predictive performance on the test dataset.

(Insert Confusion Matrix Here)

The confusion matrix indicates that the selected model correctly classified the majority of patients with very few misclassifications.

(Insert Classification Report Here)

The classification report shows high precision, recall, and F1-score for both classes, confirming the reliability of the model.

CHAPTER 10 CONCLUSION

This project successfully developed a Heart Attack Prediction System using machine learning techniques. The dataset was pre-processed and analysed using various statistical and visualization methods. Exploratory Data Analysis provided insights into patient characteristics and feature relationships. Multiple classification algorithms were trained and compared. The Decision Tree and Random Forest models achieved the highest accuracy of approximately **99%**, demonstrating excellent predictive capability. The developed system can assist healthcare professionals in the early detection of heart disease and support clinical decision-making.

FUTURE SCOPE

The project can be extended in several ways:

- Integrate real-time patient data from hospitals.
- Develop a web application for online prediction.
- Deploy the model using Streamlit or Flask.
- Increase dataset size to improve generalization.
- Apply deep learning techniques for enhanced performance.
- Integrate wearable health device data for continuous monitoring.

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